👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 1 Measuring Tools (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 1: Measurement Lab Sheet (dial calipers, micrometer, stations), Micrometer Handout, Measuring cylinder bore video”)	Level 2 piece
Measurement Lab Sheet (4 stations)	Measurement Stations lab packet + notes A, B
Micrometer Handout (scanned PDF)	Slides “Reading a Micrometer” + notes B
Measuring cylinder bore video	Video guide + slides “Measuring the Cylinder Bore”
Class slides	Deck: 16 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 10 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Digital calipers (if available) for students who cannot read a dial; same points
- Calculator for adding bar + dial
- Number line card for thousandths

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Measurement Lab: L1 picks the right tool and an adult reads the number; L2 reads a digital or dial caliper; L3 reads a micrometer with a frame
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Micrometer Handout**: The handout is a scanned image with no text; our notes teach the same skill.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Measurement Stations (Level 2)

Small Engines Overhaul · Unit 1 · goes with the class “Measurement Lab Sheet” (lab points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Measurement Lab Sheet”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (lab points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Station specs depend on the engines in Mr. Davis's shop (the Lab Sheet leaves the model blank); staff fill in the spec column.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

📏 Measuring Tools Notes

Small Engines Overhaul · Unit 1 Measuring Tools · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Tools

1. Match the tool to its job:

Ruler	very exact outside size
Dial caliper	small gaps
Micrometer	length to 1/16 inch
Feeler gauge	outside, inside and depth

B. Reading tools

2. Caliper: add the bar and the _____.
3. Micrometer: add the sleeve and the _____.
4. One _____ of an inch is 0.001.

Word bank: **dial** **thimble** **thousandth**

C. Specs

5. The repair manual gives the _____.
6. Smaller than spec means the part is _____.
7. Measure the cylinder at the top, middle and _____.

Word bank: **bottom** **specification** **worn**

Level 2 version of the gen ed “Measurement Lab Sheet + Micrometer Handout.”

📖 Measuring Tools: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Ruler

A straight tool marked in inches or millimeters to measure length.



Dial caliper

A sliding tool with a dial that measures outside, inside and depth to 0.001 inch.



Micrometer

A screw tool that measures very exactly, to one thousandth of an inch (0.001).



Feeler gauge

A set of thin metal blades of known thickness used to measure small gaps.



Specification

The correct size or number the maker says a part should be (the 'spec').



Thousandth

One thousandth of an inch, written 0.001. Engine parts are measured this small.



Diameter

The distance straight across a circle through the middle, like across a crankshaft journal.



Cylinder bore

The round hole in the engine block where the piston goes up and down.

📏 Measurement Stations

Small Engines Overhaul · Unit 1 · Level 2



Name: _____

Modified from the class “Measurement Lab Sheet”.

🧺 Supplies

- ☐ 🗜️ **Dial caliper** or digital
- ☐ 🔩 **Micrometer** 0-1 inch
- ☐ 📄 **Feeler gauge set**
- ☐ 📏 **Ruler**
- ☐ 🔧 **Rod and crankshaft** station parts
- ☐ 🧽 **Clean rag**

📏 Steps: check each box when you finish

1. 🧽 Wipe each station part clean. ☐
2. 🗜️ Station 1: measure the rod diameter and length with the caliper. ☐
3. 🔩 Station 2: measure the crankshaft journals with the micrometer. ☐
4. 📄 Station 3: find the feeler blade that fits the sample gap. ☐
5. 📏 Station 4: measure a part with a ruler to the nearest 1/16. ☐
6. 🎯 Write measured and spec side by side. ☐
7. 📦 Return tools to their cases. ☐

🚧 **Under construction:** Station specs depend on the engines in Mr. Davis's shop (the Lab Sheet leaves the model blank); staff fill in the spec column. *Check with the class teacher.*

📏 Measurement Stations · Prep Sheet

Small Engines Overhaul · Unit 1 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 🗜️ **Dial caliper** or digital
- ☐ 🔩 **Micrometer** 0-1 inch
- ☐ 📄 **Feeler gauge set**
- ☐ 📏 **Ruler**
- ☐ 🔧 **Rod and crankshaft** station parts
- ☐ 🧽 **Clean rag**

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

💔 Measuring tools are **delicate**: no dropping, no forcing.

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

📏 Measurement Stations · Reflection

Small Engines Overhaul · Unit 1 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Measurement Stations steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Measuring Tools Quiz

Small Engines Overhaul · Unit 1 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🗜️ A dial caliper can measure…

A. Only weight | B. Outside, inside and depth | C. Only temperature

2. 🔩 A micrometer measures to…

A. 1 inch | B. 1 foot | C. 0.001 inch

3. 🔢 0.001 is read as…

A. One thousandth | B. One hundred | C. One tenth

4. 🗜️ Bar 0.400 + dial 0.020 =

A. 0.600 | B. 4.20 | C. 0.420

5. 📄 A feeler gauge measures…

A. Small gaps | B. Engine weight | C. Oil level

6. 🎯 The spec is…

A. Your guess | B. The correct size from the manual | C. A tool

7. 📉 Measured smaller than spec means…

A. The part is new | B. The part is worn | C. Nothing

8. 🕳️ The cylinder bore is measured…

A. At top, middle and bottom | B. Only once | C. With a ruler only

9. 🔄 Close a micrometer using the…

A. Hammer | B. Pliers | C. Ratchet, gently

10. 🧽 Before measuring, the part should be…

A. Clean | B. Oily | C. Painted

🎬 Measuring Cylinder Bore

Small Engines Overhaul · Unit 1 · Level 2 video guide (modified). YouTube video linked in Canvas: https://www.youtube.com/watch?v=6jsiVBDfqz4

Name: _____

Before you watch

What part moves up and down inside the cylinder?

_____

While you watch: circle the best answer

1. What is the cylinder bore?

A. The gas tank | B. The hole the piston moves in | C. The muffler

2. Where is the bore measured?

A. Only the top | B. Only outside | C. Top, middle and bottom

3. Which tools are used?

A. Bore gauge and micrometer | B. Ruler and tape | C. Screwdriver

4. Why measure in two directions?

A. To save time | B. No reason | C. To find out-of-round wear

After you watch: finish the sentence

5. After watching, finish the sentence:

A worn cylinder is a problem because _____

Goes with the class “Measuring cylinder bore (video notes)” (class points pts). Same points when complete.

📏 Measuring Tools Notes · ANSWER KEY

Small Engines Overhaul · Unit 1 Measuring Tools · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Tools

1. Match the tool to its job:

(Key: Ruler → length to 1/16 inch · Dial caliper → outside, inside and depth · Micrometer → very exact outside size · Feeler gauge → small gaps)

Ruler	very exact outside size
Dial caliper	small gaps
Micrometer	length to 1/16 inch
Feeler gauge	outside, inside and depth

B. Reading tools

2. Caliper: add the bar and the **dial**.
3. Micrometer: add the sleeve and the **thimble**.
4. One **thousandth** of an inch is 0.001.

Word bank: **dial** **thimble** **thousandth**

C. Specs

5. The repair manual gives the **specification**.
6. Smaller than spec means the part is **worn**.
7. Measure the cylinder at the top, middle and **bottom**.

Word bank: **bottom** **specification** **worn**

Level 2 version of the gen ed “Measurement Lab Sheet + Micrometer Handout.”
🔑 Key written by our team from the class teacher's materials, please check.

✅ Measuring Tools Quiz · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🗜️ A dial caliper can measure… Station 1

A. Only weight | **B. Outside, inside and depth** | C. Only temperature

2. 🔩 A micrometer measures to… Station 2

A. 1 inch | B. 1 foot | **C. 0.001 inch**

3. 🔢 0.001 is read as… Notes

A. One thousandth | B. One hundred | C. One tenth

4. 🗜️ Bar 0.400 + dial 0.020 = Notes

A. 0.600 | B. 4.20 | **C. 0.420**

5. 📄 A feeler gauge measures… Feeler gauge

A. Small gaps | B. Engine weight | C. Oil level

6. 🎯 The spec is… Notes

A. Your guess | **B. The correct size from the manual** | C. A tool

7. 📉 Measured smaller than spec means… Notes

A. The part is new | **B. The part is worn** | C. Nothing

8. 🕳️ The cylinder bore is measured… Bore video

A. At top, middle and bottom | B. Only once | C. With a ruler only

9. 🔄 Close a micrometer using the… Micrometer Handout

A. Hammer | B. Pliers | **C. Ratchet, gently**

10. 🧽 Before measuring, the part should be… Notes

A. Clean | B. Oily | C. Painted

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Measurement Lab Sheet + Micrometer Handout”. Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.

🎬 Measuring Cylinder Bore · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 2 video guide (modified). YouTube video linked in Canvas: https://www.youtube.com/watch?v=6jsiVBDfqz4

Name: _____

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3. Which tools are used?

A. Bore gauge and micrometer	B. Ruler and tape	C. Screwdriver

4. Why measure in two directions?

A. To save time	B. No reason	**C. To find out-of-round wear**

After you watch: finish the sentence

5. After watching, finish the sentence:

A worn cylinder is a problem because **the piston rings cannot seal, so compression drops**

Goes with the class “Measuring cylinder bore (video notes)” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.